BIHAR ENGINEERING UNIVERSITY, PATNA

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

B.Tech 4th Semester • Examination Resource

DIGITAL CIRCUITS

Exam Date: Tuesday, 3 Nov 2026 (2:00 PM - 5:00 PM)

Included Question Papers

Year	Paper Title	Course Code	Examination Session
2023	Digital Circuits	104403	End Semester Examination - 2023
2024	Digital Electronics	100403	B.Tech 4th Semester Examination, 2024
2023 (Special)	Digital Electronics	100403	Special Examination - 2023
2023 (Regular)	Digital Electronics	100403	End Semester Examination - 2023

Apna BEU Question Bank Archive • Compiled for B.Tech ECE 4th Semester

Contains official previous year questions with multiple-choice questions, descriptive problems, and complete circuits.

BIHAR ENGINEERING UNIVERSITY, PATNA

End Semester Examination - 2023

DIGITAL CIRCUITS

Time: 03 Hours | Code: 104403 | Full Marks: 70

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Write the short answer of the following (Any seven question only): **[14]**

1. Find the hexadecimal equivalent of the decimal number 256.
2. What is meant by weighted and non-weighted coding?
3. State Demorgan's theorem.
4. What do you mean by carry propagation delay?
5. What do you mean by Encoder?
6. What are the primary disadvantages of an asynchronous counter?
7. What are races and cycles?
8. Why is MUX called as data selector?
9. When can RTL be used to represent digital system?
10. Draw the DTL based NAND gate.

Q.2 Solve both questions : **[10+4=14]**

1. Describe the procedure to design of asynchronous sequential circuit in detail with an example.
2. What are the disadvantages of asynchronous sequential circuit?

Q.3 Solve both questions : **[10+4=14]**

1. Explain the characteristics and about TTL parameters. Discuss implementation of the following digital logic families : (a) CMOS (b) TTL
2. Discuss about TTL parameters.

Q.4 Solve both questions : **[7+7=14]**

1. Explain the realization of JK flipflop from T flipflop.
2. Design a 4-bit BCD ripple counters.

Q.5 Solve this question : **[14]**

1. What is meant by magnitude comparator? Discuss in detail about two bit magnitude comparator.
2. Design a full adder using NOR gates.

Q.6 Solve both questions : **[7+7=14]**

1. Explain briefly about SOP and POS forms with example.
2. Plot the logical expression $ABCD + A\overline{B}\overline{C}\overline{D} + A\overline{B}C + AB$ on a 4 variable K-map. Obtain the simplified expression from the map.

Q.7 Solve both questions : **[7+7=14]**

1. How does don't care condition in K-map help for circuit simplification?
2. Write the POS form of the SOP expression $f(x, y, z) = \overline{X}YZ + XY\overline{Z} + X\overline{Y}Z$.

Q.8 Solve both questions : **[7+7=14]**

1. Implement the following function using suitable multiplexer. $F(a, b, c) = \Sigma m(3, 4, 5, 7)$.
2. Implement JK flipflop using SR flipflop.

Q.9 Write Short notes on any two of the following: **[14]**

1. (a) Analog to Digital Converter
2. (b) Multivibrator
3. (c) Monostable and Astable timer

BIHAR ENGINEERING UNIVERSITY, PATNA

B.Tech 4th Semester Examination, 2024

DIGITAL ELECTRONICS

Time: 03 Hours | Code: 100403 | Full Marks: 70

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Answer of the following questions (any seven only): **[14]**

1. What is the use of K-map?
2. Discuss Universal gates.
3. Add hexadecimal number 2ABC & 98F2.
4. Number of 2:1 mux requires designing 256:1 mux is .....
5. Find 2's complement of 1011011.
6. What is the function of a sample-and-hold circuit?
7. Define term propagation delay.
8. Define race around condition in JK flip flop.
9. What is the purpose of expanding memory size?
10. Differentiate between ROM and RAM.

Q.2 Solve both questions : **[14]**

1. Realize XNOR logic function using NAND gate only.
2. Simplify $Y = ABC + AB\overline{C} + A\overline{B}C$

Q.3 Solve this question : **[14]**

1. A logic circuit has four inputs A, B, C, D and output Y. Y = 1 when A & B are both 1 subjected to the condition that C and D are both low or both high. Design the logic circuit.

Q.4 Solve both questions : **[14]**

1. Design 8 to 3 line Encoder circuit.

2. Implement NAND gate using TTL logic family.

Q.5 Solve both questions : **[14]**

1. Design a parallel-to-serial converter using shift registers and explain its operation.

2. Compare the characteristics and applications of JK and T flip-flops.

Q.6 Solve both questions : **[14]**

1. Explain the working principle of an R-2R ladder DAC with a detailed diagram.

2. Design a 3-bit flash ADC and explain its working with an example.

Q.7 Solve both questions : **[14]**

1. Find the Simplified logical expression for Y.
   $Y(A, B, C, D, E) = \Sigma m$(0, 2, 4, 7, 8, 10, 12, 16, 18, 20, 23, 24, 25, 26, 27, 28)

2. Summarize the design procedure for a synchronous sequential circuit.

Q.8 Solve both questions : **[14]**

1. Implement S-R, T, D flip-flops using J-K flip-flop. Also show the implementation with help of State Tables.

2. Discuss the organization and operation of content-addressable memory (CAM).

Q.9 Write short notes on the following: **[14]**

1. (a) Binary Parallel Adder
   (b) Digital IC logic families

BIHAR ENGINEERING UNIVERSITY, PATNA

Special Examination - 2023

DIGITAL ELECTRONICS

Time: 03 Hours | Code: 100403 | Full Marks: 70

Instructions:

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2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Answer any seven question only: [14]

1. Define Dynamic RAM.
2. The resolution of a 10-bit AD converter for an input range of 10 V is approximately _____
3. The parameter through which 16 distinct values can be represented is known as _____
4. Define Multiplexing.
5. Write the significance of Truth table.
6. Write the differences between Combinational and Sequential circuit.
7. A hexadecimal odometer displays F52F. The next reading will be _____
8. The basic property of Sequential circuit is _____
9. How many entries will be in the truth table of a 4-input NAND gate?
10. What is Dual Slope ADC?

Q.2 Solve this question : [14]

1. What are weighted, non-weighted, cyclic and self-complementary codes? Explain each with examples.

Q.3 Solve this question : [14]

1. Design a BCD to 7-segment display decoder circuit using logic gates.

Q.4 Solve this question : [14]

1. Design a 3-bit parallel comparator A/D convertor that provides output in 2's complement format.

Q.5 Solve both questions : **[14]**

1. Design full adder using 8:1 mux.

2. Design full adder using 4:1 mux.

Q.6 Solve this question : **[14]**

1. What is binary shift register? Write down their application.

Q.7 Solve this question : **[14]**

1. Minimize the following expression using k-map
   $f(P, Q, R, S) = \Sigma m(0, 1, 4, 5, 7, 12, 13)$

Q.8 Solve this question : **[14]**

1. Describe the procedure to design Mod-6 counter.

Q.9 Write short notes on any two of the following: **[14]**

1. i. RAM
   ii. ROM
   iii. CMOS logic
   iv. Operation of TTL logic circuit working as NAND Gate

BIHAR ENGINEERING UNIVERSITY, PATNA

End Semester Examination - 2023

DIGITAL ELECTRONICS

Time: 03 Hours | Code: 100403 | Full Marks: 70

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Answer any seven Question only: [14]

1. Draw Truth table of JK flip flop.
2. Convert the Decimal number (108.025) to their binary equivalent.
3. Draw the truth table and logic circuit of Half-adder.
4. Write the difference between combinational & Sequential circuit.
5. Draw timing diagram of SR flip-flop.
6. Find 2's complement of 1011011.
7. Number of 2:1 mux requires designing 256:1 mux is
8. Add hexadecimal number 2ABC & 98F2.
9. Discuss Universal gates.
10. What is the use of K-map?

Q.2 Solve this question : [14]

1. Implement the following function using only NAND gate $F(A, B, C) = \Sigma m(0, 1, 2, 3, 7)$.

Q.3 Solve both questions : [7+7=14]

1. Realize XNOR logic function using NAND gate only.
2. Simplify $Y = ABC + AB\overline{C} + A\overline{B}C$

Q.4 Solve this question : [14]

1. A logic circuit has four inputs A,B,C,D and output Y. Y=1 when A & B are both 1, subjected to the condition that C and D are both low or both high. Design the logic circuit.

Q.5 Solve this question : **[14]**

1. Explain Master-slave flip flop. What are race around condition?. How it can be circumvented with the help of Master-slave flip flop.

Q.6 Solve both questions : **[7+7=14]**

1. Design 8 to 3 line Encoder circuit.
2. Implement NAND gate using TTL logic family.

Q.7 Solve this question : **[14]**

1. Summarize the design procedure for a synchronous sequential circuit.

Q.8 Solve both questions : **[7+7=14]**

1. Explain the working of R-2R Ladder DAC.
2. Design 3-bit binary counter using T flip-flop.

Q.9 Write short notes on the following: **[14]**

1. (a) Binary Parallel Adder
2. (b) Digital IC logic families